

Cymbal Copper Pty Ltd (fictional) Ridgeback Concentrator Project RB-4410	Ball Mill - Process Data Sheet Equipment tag: ML-3201	Doc No. RB-4410-PS-ML3201	Rev B
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General (cover sheet)

Parameter	Value
Service	Ball Mill
Plant area	Area 32 - Ball Milling & Classification
Equipment class	Grinding Mill
Function	Secondary grinding in closed circuit with cyclone cluster CY-3201.
Connected equipment	CY-3201, PP-3201A

Process design data

Parameter	Value
Diameter x EGL	8.2 x 13.4 (27 ft x 44 ft) m
Installed power	22,000 kW
Drive	Gearless motor drive (GMD), fixed speed
Speed	75 % critical
Ball charge	32 - 35 % vol
Media size	65 mm
Specific energy	8.0 kWh/t

Revision history

Rev	Date	Description
A	2025-09-12	Issued for review
B	2026-02-20	Issued for design

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Mechanical design

Parameter	Value
Shell material	ASTM A516 Gr 70
Shell thickness	80 mm
Head/trunnion	Cast steel, one-piece
Feed chute	Rubber-lined, replaceable wear plates
Discharge	Overflow to PP-3201 hopper
Reline method	Mill relining machine, 2.5 t capacity
Inching drive	GMD inching mode, 0.1 rpm
Charge weight at max filling	1,980 t

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Drive, motor and lubrication

Parameter	Value
Motor type	Ring motor (GMD)
Rated motor power	20,500 kW
Voltage	33 kV feed / cycloconverter
Lube system	Combined hydrostatic lift / hydrodynamic lube unit
HP lift pump discharge pressure (normal)	110 bar(g)
Lube system design pressure	170 bar(g)
Bearing temperature instruments	TT-3201 / TT-3202
Oil grade	ISO VG 460